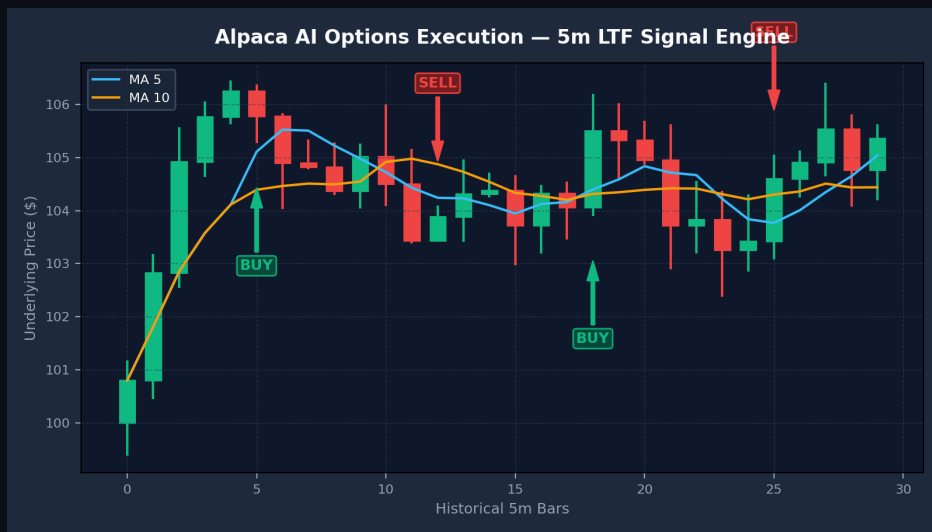


AXE Genesis — Alpaca AI Options RL Pipeline

DXY Divergence Alpha, Multi-Horizon Meta-Learner & Contextual Q-Learning Execution



Core Strategy: Exploiting Dollar Index (DXY) vs Asset divergence at key SNR zones.

Two-Tier Framework: HTF Meta-Learner (Tier 1) + Dual-Branch Q-Executor (Tier 2).

Public Repository: <https://github.com/SirDaniell/alpaca-ai-trading-agent>

Framework Parity: 1:1 functional PyTorch & Keras reinforcement learning models.

Reward Shaping: DeepScalper hindsight bonus, 'Wise Patience' & zero-chase action masking.

Autonomous API/CLI: Direct execution via Alpaca Trading API & Alpaca CLI wrapper.

1. Primary Trading Edge: DXY Divergence & Zone Anchoring

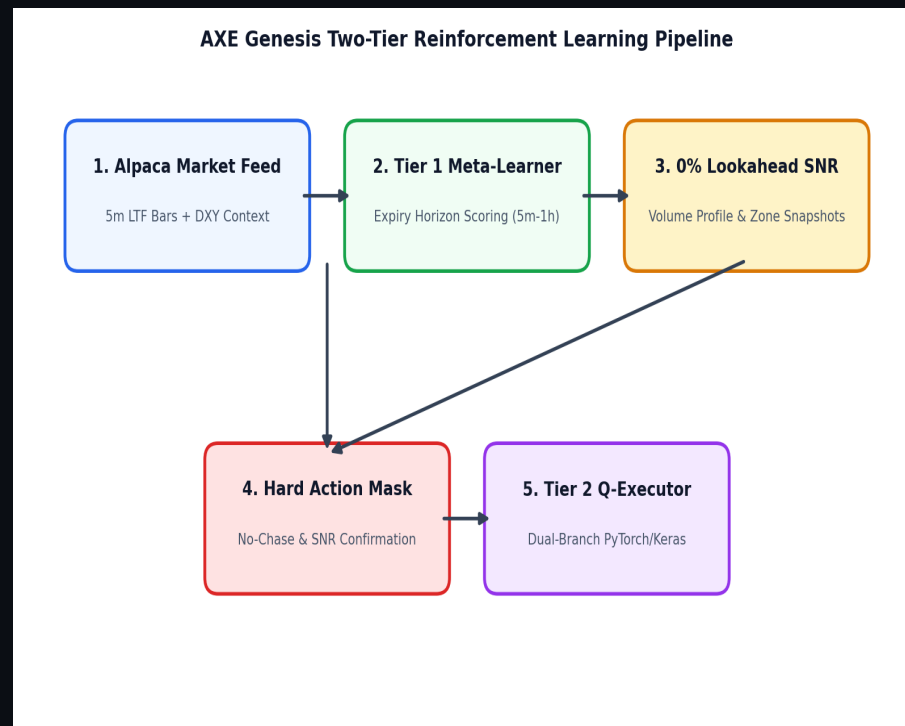
Capturing structural momentum shifts when underlying asset price diverges from the US Dollar Index.

Dollar Index (DXY) Macro Divergence:

- Tracks real-time inverse correlation between DXY basket (EURUSD, USDJPY, GBPUSD, USDCAD) and traded underlying.
- Identifies bullish divergence (Asset making higher low while DXY fails to break higher) and bearish divergence.
- Divergence signals act as high-probability macro catalysts to filter out false breakouts.

Zone-Anchored No-Chase Execution:

- Causal 0%-lookahead Support & Resistance detection calculates real-time zone boundaries.
- HardActionMask strictly prohibits BUY_CALL / BUY_PUT unless price is within ATR proximity of a confirmed zone.
- Volume profile delta (buyers vs sellers) validates breakout/reversal before unmasking entry actions.



2. Two-Tier Reinforcement Learning Architecture

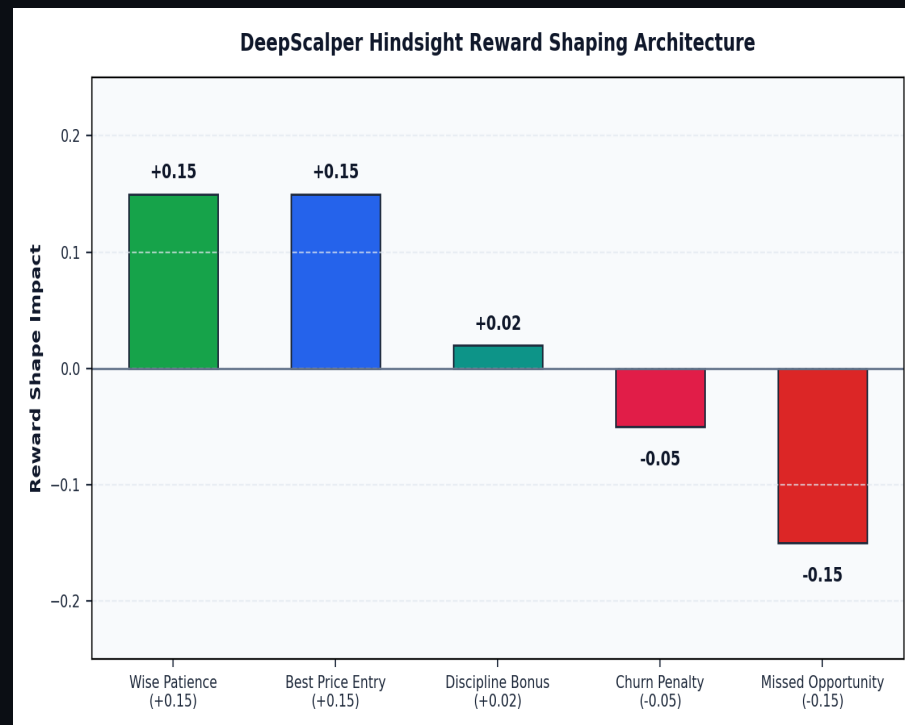
Decoupled multi-horizon context scoring and zero-interference execution.

Tier 1 — Signal Meta-Learner:

- Evaluates forward price statistics (MFE, MAE, reversal timing) to output conviction score [0.0 - 1.0].
- Predicts optimal option expiry duration (5m, 15m, 30m, 1h) based on MTF RSI & DXY regime context.
- Features 6 decoupled auxiliary regression heads with private projections for zero gradient interference.

Tier 2 — Dual-Branch Ensemble Options Q-Learner:

- Microstructure + MTF Alignment gated fusion layer.
- 5-action space: WAIT, CALL, PUT, TP_HALF, CLOSE.
- Full 1:1 functional parity between PyTorch and Keras.



3. DeepScalper Reward Shaping & Data Integrity

Aligning execution discipline against opportunity cost with zero-lookahead guarantees.

DeepScalper Hindsight Reward Shaping:

- **Wise Patience Bonus (+0.15):** Rewards WAIT when standing down avoids adverse drawdown.
- **Missed Opportunity (-0.15):** Penalizes WAIT if an unmasked setup played out successfully.
- **Best Price Entry (+0.15):** Rewards entries with minimal drawdown exposure ($\leq 0.5\%$).

Zero-Lookahead Causal SNR Zone Detection:

- Vectorized `detect_snr_levels_sequential` ensures zone snapshots are generated strictly on `df.iloc[:idx+1]`.
- Completely eliminates lookahead leakage across historical training, Kaggle GPU pipelines, and live loops.

Dynamic Instrument Scaling:

- `get_instrument_metadata` automatically maps Equities/ETFs (SPY, QQQ, GLD, AAPL) to \$0.01 = 1 cent = 1 pip.
- Eliminates 100x scale distortions between Equities and FX/Crypto.

4. Cross-Symbol Out-of-Sample Benchmark

Multi-symbol walk-forward backtesting across standardized option underlyings.

Cross-Symbol Benchmark Loop:

- Evaluates 40,000 historical 5m bars (~2 years of market data) per symbol across ETF/Equity universe.
- Out-of-sample holdout test set (20% split, ~8,000 bars) walk-forward execution.

Multi-Head Loss Convergence:

- Multi-task auxiliary heads (pips, risk, liquidity, reversal) use isolated gradient paths (`feat.detach()`).
- Prevents auxiliary loss spikes from corrupting primary directional Q-learning policies.

Expiry Duration Optimization:

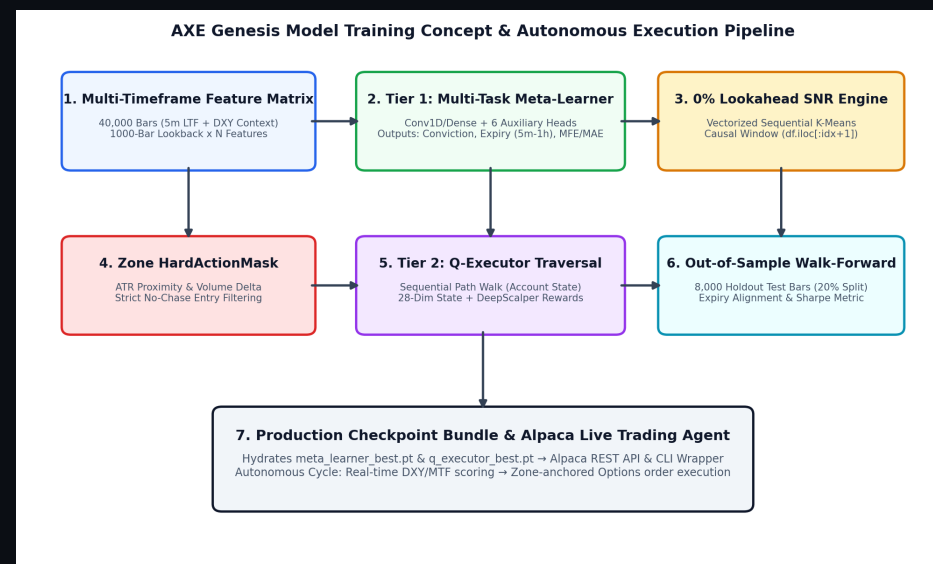
- Dynamically selects contract expiries across 5m (1 bar), 15m (3 bars), 30m (6 bars), and 1h (12 bars).
- Maximizes Sharpe ratio and win rate by matching expiry to predicted trend momentum.

5. Model Training Concept & Pipeline Workflow

4-Phase Kaggle GPU Pipeline: Multi-Task Meta-Learning + Sequential Q-Traversal.

Training Concept Highlights:

- **Phase 1 Multi-Task Meta-Learner:** 50 epochs over 40,000 bars using Cosine Annealing ($1e-3 \rightarrow 1e-5$). Decoupled auxiliary heads eliminate gradient interference.
- **Phase 2 Sequential Traversal:** Sequential walk preserves path-dependent account state (drawdown %, streaks, open positions).
- **Phase 3 & 4 Out-of-Sample Eval:** Walk-forward backtest on 20% holdout bars (~8,000 bars) across 5m, 15m, 30m, 1h options.
- **Phase 5 Hydration Checkpoints:** Exports meta_learner_best.pt and q_executor_best.pt directly into backend engine.



6. Summary & Hackathon Highlights

Production-grade autonomous options trading agent built on Alpaca infrastructure.

Key Feature	Implementation Details
Primary Alpha Edge	DXY vs Underlying divergence combined with zero-lookahead SNR zone anchoring.
Two-Tier RL Architecture	Tier 1 Meta-Learner (Expiry & Conviction) + Tier 2 Dual-Branch Q-Executor (Entry & Risk).
Alpaca Integration	Full Alpaca Trading API REST client + Alpaca CLI wrapper for order & position tracking.
Public Repository	https://github.com/SirDaniell/alpaca-ai-trading-agent
PyTorch / Keras Parity	Complete dual-framework implementation for maximum research flexibility.

Submitted for lablab.ai Alpaca AI Hackathon — Built with PyTorch, Keras, Alpaca API & CLI.